Module Code	Examiner	Academic Unit	Tel
CPT208		Computing	

Xi'an Jiaotong-Liverpool University
西交利物浦大学

2^{nd} Semester, 2024/25 Final Examination

Human-Centric Computing

Time Allowed: 2 Hours

Instructions to Candidates

1. This is an open-book examination with invigilation. You are allowed to take one-page notes with you. Please complete the assessment independently and honestly.
2. Total marks available are **100**.
3. This exam consists of **three questions**. The mark allocated for each question is indicated at the end of the question.
4. Answer **ALL** questions. There is **NO** penalty for providing a wrong answer.
5. Only **English** solutions are accepted. Answer should be written in the answer booklet provided. Clearly indicate the **question numbers** before your solutions. Submit your notes together with your answer booklet. Make sure your writings are clear to read.
6. All materials must be returned to the exam supervisor upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.

Question 1 Discovering Requirements and Design (40 marks)

Imagine that you are tasked with designing a *university digital services application* that enables students to **manage their academic activities**, including course registration, accessing learning resources, and scheduling appointments with faculty. You must adopt a user-centered design approach. Based on the given context, answer the questions below. Note that your answers should be closely related to the given context.

(a) As two main types of goal in user-centered design, what is the difference between usability and user experience? Then, please list three design goals for **both** usability and user experience in this application, respectively. [10]

(b) Besides the usability and user experience, please outline the four other common types of requirements in human-centered design. Then, please choose one of them as an example to explain why it is important and how it will affect the design of this application. [10]

(c) If designers decide to keep the same order of items in the application's menu, which of Shneiderman's Eight Golden Rules does this decision reflect? Next, please identify the design principle this rule belongs to and provide its definition, then give another example of how this principle could be applied in the design of this application. [10]

(d) If designers integrate XIPU AI (a chatbot) into the application, what type of interaction does this represent? Next, please discuss the pros and cons of this interaction type, and list examples of how these advantages and disadvantages might apply to the designed application. [10]

Question 2 Prototype (30 marks)

Imagine you are part of a design team creating *a mobile application to support student well-being on campus.* The app includes features such as mood tracking, access to mental health resources, booking counseling sessions, and sending reminders for healthy habits (e.g., hydration, breaks, sleep). Your team is now moving into the prototyping phase.

(a) During the initial design phase, your group chooses to use prototypes to evaluate user feelings. Please address which prototyping aim does this approach represent, and what are its key benefits? Then, please suggest which tool would you select for this purpose, and how would you use it to achieve your goal? [10]

(b) Please list the five common dimensions of prototyping fidelity. Next, please select the most important dimension for this application's design, define it, and explain how you would determine its fidelity level, with examples. [10]

(c) The application's key function is to provide advice to students under pressure. To design this feature, please select a prototyping tool/method and briefly introduce it. Then, discuss its pros and cons for this use case, and outline your prototyping process — how would you design and test it? [10]

Question 3 Evaluation (30 marks)

You are working on a new educational app called *LearnSmart* aimed at improving math skills for high school students. The company wants to evaluate the effectiveness of *LearnSmart* compared to a traditional classroom setting. You are tasked with designing a systematic evaluation using experimental design.

(a) How can you apply quantitative and qualitative methods in the evaluation? Name a typical example for each and compare and contrast the differences. Give specific details to explain how the two methods can be used in the given context. [15]

(b) Your team plans to conduct an experimental study to evaluate if *LearnSmart* leads to better math skills and higher student satisfaction compared to traditional classroom instruction. Identify the independent variable(s) and dependent variable(s) in this experiment. What are the null hypotheses? Provide a detailed experimental plan that describes your preparations, the experimental settings, what students need to do, how to collect, analyze, and interpret the data. [15]

——— *End of Questions* ———